

5. Dielectrics

1) Electric dipole and dipole moment

- ① An electric dipole is formed by two closely placed charges, separated at a small distance d .

The two charges have opposite polarity but the same charge amount of q . The field lines are quite complicated (refer to the homework) and seemingly form loops.

- ② Far away from the dipole, the potential decays faster than $\frac{1}{r}$. More quantitatively,

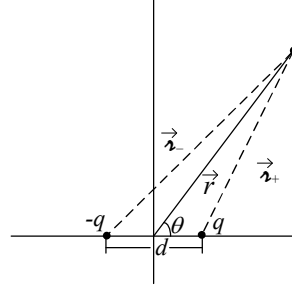


Figure. Illustration of the dipole.

$$r_+ = r - \frac{d}{2} \cos \theta = r \left(1 - \frac{d}{2r} \cos \theta \right)$$

$$\frac{1}{r_+} = \frac{1}{r} \left(1 - \frac{d}{2r} \cos \theta \right)^{-1}$$

$$\therefore \frac{1}{1-x} = 1 + x + x^2 + \dots$$

$$\therefore \frac{1}{r_+} \approx \frac{1}{r} \left(1 + \frac{d}{2r} \cos \theta \right)$$

$$\text{Similarly, } \frac{1}{r_-} \approx \frac{1}{r} \left(1 - \frac{d}{2r} \cos \theta \right)$$

$$\text{As a result, } \frac{1}{r_+} - \frac{1}{r_-} \approx \frac{d}{r^2} \cos \theta$$

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_+} - \frac{q}{r_-} \right) \approx \frac{q}{4\pi\epsilon_0} \frac{d \cos \theta}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$$

where $\vec{p} = q\vec{d}$ ($\vec{d}$: pointing from $-q$ to $+q$) is defined as the dipole moment.

Now, label the positions of the two charges with $\vec{r}'_1$ and $\vec{r}'_2$, $\vec{d} = \vec{r}'_2 - \vec{r}'_1$, as shown below.

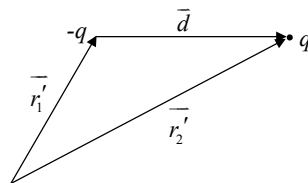


Figure. Denote the positions of the two opposite charges with $\vec{r}'_1$ and $\vec{r}'_2$.

We found that, the dipole moment can be written as:

$$\vec{p} = q\vec{d} = q(\vec{r}'_2 - \vec{r}'_1) = q\vec{r}'_2 + (-q)\vec{r}'_1$$

For a continuous distribution of charges, the dipole moment can be generalized from the above expression into

$$\vec{p} = \int_V \vec{r}' \rho(\vec{r}') dV'$$

And the corresponding potential can still be calculated using

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$$

2) Polarization of a dielectric material

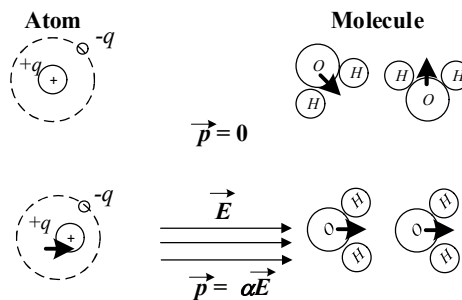


Figure. After an external electric field is applied, atoms and molecules tend to acquire a net dipole moment.

For insulators, the lack of free charges does not mean that they do not respond to externally applied electric fields. The electric field tends to induce dipole moments in the atoms and molecules of the material. Such microscopic dipole moments, when superimposed, manifests as macroscopic polarization of the material. Microscopically, the induced dipole moment $\vec{p}$ is linearly related to the local electric field $\vec{E}$:

$$\vec{p} = \alpha \vec{E}$$

Here the proportional factor, α , is called the polarizability. In general, it is a tensor. For the simplest case that we consider in this course, α is a scalar constant, meaning that $\vec{p}$ is always parallel with $\vec{E}$.

Like the concept of charge density, which describes the amount of net charge within a unit volume, we define the dipole moment density as $\vec{P} = \frac{\text{total dipole moment within } V}{\text{volume } V}$. This parameter

is termed “polarization”, whose unit is: $[\vec{P}] = \frac{[p]}{[m^3]} = \frac{[c][m]}{[m^3]} = \frac{[C]}{[m^2]}$, which is the same as surface charge density. Note that we use capital letter P for polarization, and use lower-case p for the dipole moment.

3) Bound charges

① Surface bound charge:

Consider a long dielectric material, uniformly polarized in the horizontal direction, see the figure below. The polarization of the material leaves a thin layer of positive charge on its right surface, while on the left surface, there is a layer of negative charge. The surface area is denoted as A .

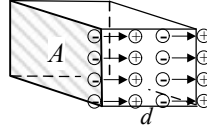


Figure. A piece of uniformly polarized dielectric material.

The whole piece can be treated as a big electric dipole, whose total dipole moment is:

$$q \vec{d} = (\sigma_b A) \cdot \vec{d} = \vec{P} (\underbrace{Ad}_{\text{volume}})$$

$\downarrow$ $\downarrow$ $\downarrow$
 Total charge on surface Surface bound charge density Density of molecular dipole

$$\therefore \boxed{\sigma_b = \vec{P} \cdot \hat{n}}$$

Note: if $\vec{P}$ and $\hat{n}$ are not parallel, σ_b will drop as a function of $\hat{P} \cdot \hat{n}$ ($=\cos \theta$) since it reflects the total “flux” of dipole moment through the surface.

② Volume bound charge:

Diverging $\vec{P} \Rightarrow$ Piling-up of negative charges inside. Consider the sphere in the following figure

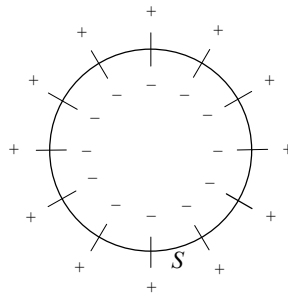


Figure. A sphere with diverging polarization.

$$\begin{aligned} \text{Total bound charge inside} &= \int_V \rho_b dV = - \underbrace{\oint_S \vec{P} \cdot d\vec{a}}_{\substack{\text{total positive} \\ \text{charge on surface}}} \stackrel{\substack{\equiv \\ \text{divergence} \\ \text{theorem}}}{=} - \int_V (\nabla \cdot \vec{P}) dV \\ \therefore \quad &\boxed{\rho_b = -\nabla \cdot \vec{P}} \end{aligned}$$

③ An alternative method to obtain the expressions for the bound charges:

Consider a polarized object, with a distribution of dipole moment density as denoted by the polarization $\vec{P}$:

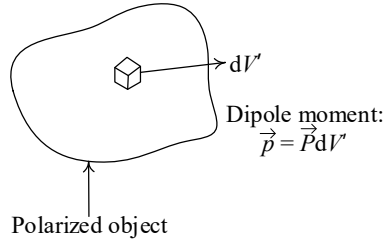


Figure. A bulk object with polarization $\vec{P}$.

Recall that a dipole, placed at the origin with a dipole moment $\vec{p}$, has a potential $V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$.

To compute the total potential of the bulk polarized object, simply apply the superposition principle, and keep in mind that the total dipole moment within volume element dV' is $\vec{P}dV'$, so

$$\begin{aligned}
 V(\vec{r}) &= \frac{1}{4\pi\epsilon_0} \int_V \frac{(\vec{P}(\vec{r}')dV') \cdot \hat{z}}{z^2} \\
 \because \nabla' \left(\frac{1}{z} \right) &= \frac{\hat{z}}{z^2} \quad (\nabla' \text{ is the same as } \nabla \text{ except it operates on } x', y', z') \\
 \therefore V(\vec{r}) &= \frac{1}{4\pi\epsilon_0} \int_V \vec{P}(\vec{r}') \cdot \nabla' \left(\frac{1}{z} \right) dV' \\
 \because \nabla' \cdot \left(\frac{\vec{P}(\vec{r}')}{z} \right) &= \vec{P}(\vec{r}') \cdot \nabla' \left(\frac{1}{z} \right) + \frac{1}{z} (\nabla' \cdot \vec{P}(\vec{r}')) \quad (\text{product rule}) \\
 \therefore V(\vec{r}) &= \frac{1}{4\pi\epsilon_0} \left[\int_V \nabla' \cdot \left(\frac{\vec{P}(\vec{r}')}{z} \right) dV' - \int_V \frac{1}{z} (\nabla' \cdot \vec{P}(\vec{r}')) dV' \right] \\
 &= \underbrace{\frac{1}{4\pi\epsilon_0} \oint_S \frac{1}{z} \vec{P}(\vec{r}') \cdot d\vec{a}'}_{(I)} + \underbrace{\frac{1}{4\pi\epsilon_0} \int_V \frac{1}{z} (-\nabla' \cdot \vec{P}(\vec{r}')) dV'}_{(II)}
 \end{aligned}$$

Compare the above expressions in (I) and (II) with the potential of surface charge distribution (a closed surface) $V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \oint_S \frac{1}{z} \sigma(\vec{r}') \cdot d\vec{a}'$, and that of volume charge distribution $V(\vec{r}) =$

$\frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}')}{z} dV'$, we readily obtain:

(I) is the potential due to surface (bound) charge density σ_b , where $\boxed{\sigma_b = \vec{P} \cdot \hat{n}}$

(II) is the potential due to surface (bound) charge density ρ_b , where $\boxed{\rho_b = -\nabla \cdot \vec{P}}$

These results are consistent with what we have concluded earlier.

4) The electric displacement

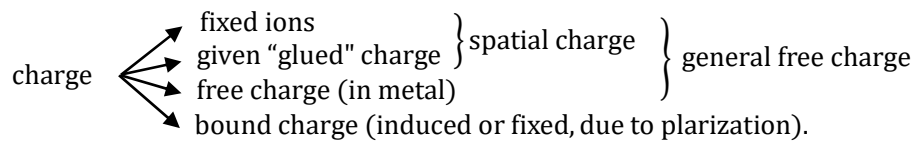
In real-world applications, computing the electric field distribution is difficult because:

- (i) Even if the free charge density ρ_f, σ_f is given, we have no idea how $\vec{P}$ (thus, ρ_b, σ_b also) is distributed, so we don't know how material polarization contributes to the total field.
- (ii) Even if $\vec{P}$ is known, calculating $\vec{E}$ is sometimes difficult.

These difficulties motivate us to introduce the electric displacement.

Before we embark on the topic, let's pay attention to the following two points:

- ① Summarize all types of charges we may encounter:



Attention: $\text{free charge} \equiv \text{all charge EXCEPT bound charge}$

- ② We emphasize one important property of the bound charge. The total bound charge (including surface and volume) should vanish since there is no net charge. This can be easily proven:

$$\begin{aligned}
 Q_{tot\ bound} &= \oint_s \sigma_b da + \int_v \rho_b dV \\
 &= \oint_s \vec{P} \cdot d\vec{a} - \int_v \nabla \cdot \vec{P} dV
 \end{aligned}$$

According to divergence theorem:

$$\oint_s \vec{P} \cdot d\vec{a} = \int_v \nabla \cdot \vec{P} dV$$

Thus, $Q_{tot\ bound} = 0$.

Now, we derive the expression for the electric displacement vector. We start from the expression of the total charge density: $\rho = \rho_b + \rho_f$, meanwhile, we invoke the Gauss's law $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$ to get:

$$\epsilon_0 \nabla \cdot \vec{E} = \rho = \rho_f + \rho_b = \underbrace{\rho_f}_{(i)} - \underbrace{\nabla \cdot \vec{P}}_{(ii)}$$

- (i) generates field I;
- (ii) generates another field (field II) to reduce the total field.

Manipulating the above equation, we get $\nabla \cdot (\epsilon_0 \vec{E} + \vec{P}) = \rho_f$, we reach the Gauss's law for the

electric displacement $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$:

$$\boxed{\nabla \cdot \vec{D} = \rho_f}$$

Integrating both sides, applying divergence theorem, we get the Gauss's law for the electric displacement in the integral form:

$$\int_V \nabla \cdot \vec{D} dV = \oint_S \vec{D} \cdot d\vec{a} = Q_{f,enc}$$

Note: There is a Gauss's law for $\vec{D}$, but no "Coulomb law" for $\vec{D}$, i.e.,

$$\vec{D}(\vec{r}) \neq \frac{1}{4\pi} \int \frac{\hat{r}}{r^2} \rho_f(\vec{r}') dV'$$

To understand why there is no Coulomb law for $\vec{D}$, recall that we have proven the following:

$$\text{If } \vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\hat{r}}{r^2} \rho_f(\vec{r}') dV' \Rightarrow \begin{cases} \nabla \cdot \vec{E} = \rho/\epsilon_0 \\ \nabla \times \vec{E} = 0 \end{cases}$$

$$\text{So, if } \vec{D} \text{ satisfies Coulomb's law, } \vec{D}(\vec{r}) = \frac{1}{4\pi} \int \frac{\hat{r}}{r^2} \rho_f(\vec{r}') dV' \Rightarrow \nabla \times \vec{D} = 0$$

$$\text{if } \nabla \times \vec{D} = 0 \Rightarrow \nabla \times \vec{D} = \nabla \times (\epsilon_0 \vec{E} + \vec{P}) = \nabla \times \vec{P} = 0 \Rightarrow \vec{P} \text{ is always conservative}$$

$\Rightarrow \vec{P}$ integrated along ANY closed loop must be zero. However, consider the configuration in the following figure:

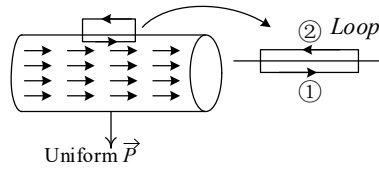


Figure. Situation in which $\oint_{loop} \vec{P} \cdot d\vec{l} \neq 0$.

The figure shows a uniformly polarized material, having a polarization $\vec{P}$ in the x direction. The loop integration shown above cannot be zero because the integration along edge ① is >0 while the integration along ② = 0 (outside the material, $\vec{P} = 0$). The contribution from the two vertical sides are effectively zero because $\vec{P}$ and $d\vec{l}$ are perpendicular on these edges. So, apparently this is in contradiction to $\nabla \times \vec{P} = 0$.

$$\text{In conclusion, } \nabla \times \vec{P} \neq 0 \Rightarrow \begin{cases} \vec{D} \text{ is not the gradient of a potential.} \\ \vec{D} \text{ does not obey Coulomb's law.} \end{cases}$$

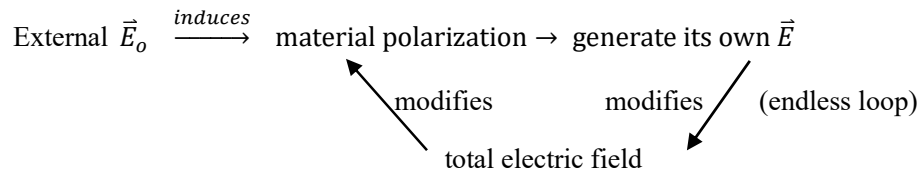
5) Linear, isotropic and homogeneous (LIH) dielectric material

Although in introduction of $\vec{D}$ can simplify computation, but in general, knowing $\vec{D}$ does not mean that $\vec{E}$ can be straightforwardly computed, because $\vec{P}$ is unknown. In most cases, $\vec{P}$ is induced by externally applied $\vec{E}$ field; if the dielectric material is linear, isotropic and homogeneous (LIH), the following relationship between $\vec{P}$ and $\vec{E}$ is valid:

$$\vec{P} = \epsilon_0 \chi_e \vec{E}$$

Note:

- ① χ_e is the electric susceptibility, which is a unit-less constant scalar.
- ② Explain LIH: linear means $\vec{P}(\vec{r}) = \epsilon_0 \vec{\chi}_e(\vec{r}) \vec{E}(\vec{r})$ where $\vec{\chi}_e$ is a spatially-varying tensor;
linear + isotropic means $\vec{P}(\vec{r}) = \epsilon_0 \chi_e(\vec{r}) \vec{E}(\vec{r})$, where $\chi_e(\vec{r})$ is a spatially dependent scalar;
linear + isotropic + homogeneous means $\vec{P}(\vec{r}) = \epsilon_0 \chi_e \vec{E}(\vec{r})$, where χ_e is a spatially-independent constant.
- ③ In $\vec{P}(\vec{r}) = \epsilon_0 \chi_e \vec{E}(\vec{r})$, $\vec{E}$ is the total field, NOT the externally applied field.
- ④ This equation greatly simplifies the computation of $\vec{E}$. Otherwise, the mathematical treatment of the polarization process can be very complex, as shown below:



- (End of note)

For a LIH material, $\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_0 \vec{E} + \epsilon_0 \chi_e \vec{E} = \epsilon_0 (1 + \chi_e) \vec{E} = \epsilon_0 \epsilon_r \vec{E} = \epsilon \vec{E}$, where:

- $\epsilon_r = 1 + \chi_e$ is the relative permittivity, sometimes it's called the "dielectric constant". For air, ϵ_r is almost 1, for SiO₂, it's 3.9, for diamond, it's 10, for water, it's ~80.
- $\epsilon = \epsilon_0 \epsilon_r$ is the electric permittivity, which directly converts $\vec{E}$ to $\vec{D}$.

Example:

For a LIH dielectric material, if ρ_f is known, compute ρ_b .

Solution:

$$\vec{P} = \vec{D} - \epsilon_0 \vec{E} = \epsilon_0 (\epsilon_r - 1) \vec{E} = \epsilon_0 (\epsilon_r - 1) \frac{\vec{D}}{\epsilon_0 \epsilon_r} = \frac{\epsilon_r - 1}{\epsilon_r} \vec{D}$$

$$\rho_b = -\nabla \cdot \vec{P} = \nabla \cdot \left(\frac{1 - \epsilon_r}{\epsilon_r} \vec{D} \right)$$

$$= \nabla \cdot \left(\frac{1 - \epsilon_r}{\epsilon_r} \right) \cdot \vec{D} + \frac{1 - \epsilon_r}{\epsilon_r} \nabla \cdot \vec{D} \quad (\text{Applying the product rule})$$

Because for the material is homogeneous, ϵ_r is a constant, thus $\nabla \cdot \left(\frac{1 - \epsilon_r}{\epsilon_r} \right) = 0$

$$\therefore \rho_b = \frac{1 - \epsilon_r}{\epsilon_r} \nabla \cdot \vec{D} = \frac{1 - \epsilon_r}{\epsilon_r} \rho_f$$

Conclusions:

In a LIH dielectric material, bound charges will always coincide with free charges.

If there is no free charge inside the material, $\rho = \rho_f + \rho_b = 0$ inside the material,

$\Rightarrow$ All net charges reside on the surface, inside the material, $\nabla^2 V = 0$

6. Boundary conditions

The electric field can be discontinuous across the boundary. To analyze the problem, we decompose the field at the boundary into two components: one perpendicular and one parallel to the boundary.

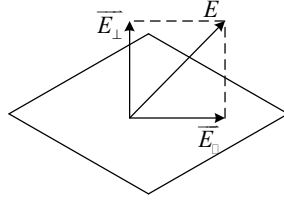


Figure. Decomposition of the electric field on the boundary.

Now, we examine the two components separately.

- 1) $\vec{E}_\perp$ (note that the direction of $\vec{E}_\perp$ is determined by the boundary, so $E_\perp$ is a scalar)

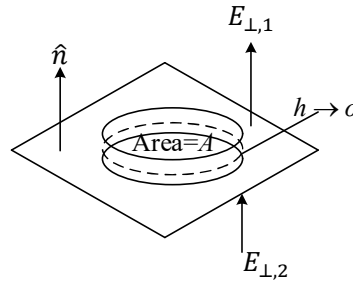


Figure. Use a cylinder across the boundary to study the perpendicular component.

Let the fields on the two sides be labeled '1' and '2', with the surface normal direction pointing from region 2 towards region 1, which is also the positive direction for both regions.

Consider a cylindrical volume across the boundary, as shown in the above figure. The discontinuity of the field can be analyzed when the height of the cylinder h approaches 0.

Within the volume, apply Gauss's law,

$$\oint \vec{E} \cdot d\vec{a} = Q_{enc}/\epsilon_0 \xrightarrow{h \rightarrow 0} E_{\perp,1}A - E_{\perp,2}A = A \cdot \sigma/\epsilon_0$$

$$\Rightarrow \boxed{E_{\perp,1} - E_{\perp,2} = \sigma/\epsilon_0}$$

- 2) $\vec{E}_\parallel$ (note that the direction of $\vec{E}_\parallel$ is within the boundary plane, so it's a vector)

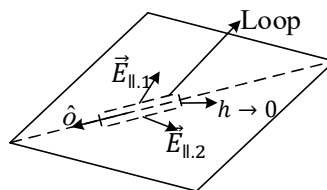


Figure. Use a rectangular integration loop to study the parallel component.

We may assume that $\vec{E}_{\parallel,1}$ and $\vec{E}_{\parallel,2}$ are unparallel in general. To study the discontinuity of the parallel component, we integrate the electric field along a rectangular loop across the boundary, as shown in the above figure. Again, we assume that the height of the loop approaches zero.

$$\oint_{Loop} \vec{E} \cdot d\vec{l} = \int_s (\nabla \times \vec{E}) \cdot d\vec{s} \equiv 0$$

Denote the direction of the integration path in region 1 as $\hat{o}$, the length of the path as l ; when h approaches 0,

$$\oint_{Loop} \vec{E} \cdot d\vec{l} = (\vec{E}_{\parallel,1} \cdot \hat{o})l + (\vec{E}_{\parallel,2} \cdot (-\hat{o}))l = \hat{o} \cdot (\vec{E}_{\parallel,1} - \vec{E}_{\parallel,2})l \equiv 0$$

Because $\hat{o}$ is an arbitrary direction in the boundary plane, the only possibility for $\hat{o} \cdot (\vec{E}_{\parallel,1} - \vec{E}_{\parallel,2}) \equiv 0$ is:

$$\boxed{\vec{E}_{\parallel,1} = \vec{E}_{\parallel,2}}$$

3) Electric potential V

$$V_1 - V_2 = \int_{region\ 1}^{region\ 2} \vec{E} \cdot d\vec{l} \xrightarrow{h \rightarrow 0} 0$$

Define $\frac{\partial V}{\partial n} = \nabla V \cdot \hat{n} = -E_{\perp}$

According to the boundary condition for $E_{\perp}$,

$$\boxed{\begin{aligned} \frac{\partial V_1}{\partial n} - \frac{\partial V_2}{\partial n} &= -\frac{\sigma}{\epsilon_0}, \\ V_1 &= V_2 \end{aligned}}$$

Note: in the above, σ is the total charge surface density, including both bound and free charge.

4) Conductor

Because $\vec{E}_{in} = 0$ in the conductor $\Rightarrow \begin{cases} E_{\perp,in} = 0 \Rightarrow E_{\perp,out} = \sigma/\epsilon_0, \\ \vec{E}_{\parallel,in} = 0 \Rightarrow \vec{E}_{\parallel,out} = 0 \end{cases}$

$$\boxed{\begin{aligned} \vec{E}_{out} &\perp \text{ surface} \\ E_{out} &= \sigma/\epsilon_0 \\ \frac{\partial V_{out}}{\partial n} &= -\sigma/\epsilon_0 \\ \sigma &= -\epsilon_0 \frac{\partial V_{out}}{\partial n} \end{aligned}}$$

5) Boundary between two LIH dielectric materials

$$\therefore \oint \vec{D} \cdot d\vec{a} = \rho_f \quad (\text{Gauss's law for } \vec{D})$$

$$\Rightarrow \boxed{D_{1,\perp} - D_{2,\perp} = \sigma_f, \quad \varepsilon_1 E_{1,\perp} - \varepsilon_2 E_{2,\perp} = \sigma_f}$$

$$\therefore \oint \nabla \times \vec{D} \cdot d\vec{a} = \int \nabla \times \vec{P} \cdot d\vec{a}$$

$$\therefore \oint \vec{D} \cdot d\vec{l} = \oint \vec{P} \cdot d\vec{l}$$

$$\Rightarrow \boxed{\vec{D}_{1,\parallel} - \vec{D}_{2,\parallel} = \vec{P}_{1,\parallel} - \vec{P}_{2,\parallel}}$$

Using the relation of electric potential and directional derivative $\frac{\partial V}{\partial n} = \nabla V \cdot \hat{n} = -E_{\perp}$, we get

$$\boxed{\begin{aligned} \varepsilon_1 \frac{\partial V_1}{\partial n} - \varepsilon_2 \frac{\partial V_2}{\partial n} &= -\sigma_f, \\ V_{out} &= V_{in} \end{aligned}}$$

7. Capacitors

1) Basics

A capacitor is an electrical component that stores and releases electrical energy by accumulating charge on two conductive plates separated by vacuum or an insulating material, as shown below.

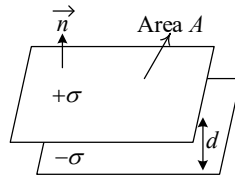


Figure. Basic structure of a parallel plate capacitor.

For simplicity, we assume that the capacitor is in a vacuum. Suppose only the top plate exists, the field discontinuity across the plate is $\frac{\sigma}{\varepsilon_0}$. Due to symmetry, the fields above and below the plate should have the same magnitude, so the field strength is

$$\Rightarrow E = \frac{\sigma}{2\varepsilon_0} \quad \left. \begin{array}{l} \uparrow \sigma/2\varepsilon_0 \\ \downarrow \sigma/2\varepsilon_0 \end{array} \right\} \text{ due to } +\sigma \text{ alone}$$

If $+\sigma$ and $-\sigma$ coexist, using the superposition principle (refer to the following figure),

$$\begin{array}{c} +\sigma \begin{array}{c} \uparrow \sigma/2\varepsilon_0 \\ \downarrow \sigma/2\varepsilon_0 \end{array} + -\sigma \begin{array}{c} \downarrow \sigma/2\varepsilon_0 \\ \uparrow \sigma/2\varepsilon_0 \end{array} \rightarrow \begin{array}{c} +\sigma \quad 0 \\ \downarrow \downarrow \downarrow \downarrow \\ -\sigma \quad 0 \end{array} \sigma/\varepsilon_0 \rightarrow \begin{array}{l} Q = \sigma A \\ \text{(on one plate)} \\ V = Ed \end{array} \end{array}$$

Figure. Illustration of computing the field distribution of a plate capacitor

We conclude that the electric field between the two plates is uniform, pointing from the plate carrying positive charge to the one carrying negative charge, with a field strength of

$$E = \frac{\sigma}{\epsilon_0}$$

The charge storage capacity of this device, at the expense of a given voltage, is defined as the capacitance C , which is found to be determined by geometric factors:

$$C = \frac{Q}{V} = \frac{Q}{E \cdot d} = \frac{\sigma A}{\frac{\sigma}{\epsilon_0} \cdot d} = \frac{\epsilon_0 A}{d} \text{ [Farad]}$$

2) Energy stored in a capacitor

Consider the situation of charging up a capacitor with a current I ,

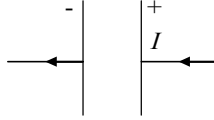


Figure. Charging a capacitor with a current.

During this process, energy is constantly flowing into the capacitor. Suppose we start to charge the capacitor when no voltage is applied (meaning that the capacitor has no pre-stored energy) until a final voltage of V_0 is reached. The current can be expressed as $I(t) = \frac{dQ(t)}{dt} = C \frac{dV(t)}{dt}$, while the instant power is $P(t) = I(t) V(t)$. Thus, the total energy dumped into the system is (don't confuse $P(t)$ with polarization $\vec{P}$)

$$U = \int_0^T P(t) dt = \int_0^T V(t) \cdot C \frac{dV(t)}{dt} dt = C \int_0^{V_0} V dV = \boxed{\frac{1}{2} C V_0^2}$$

The above result can be obtained with a different approach. Suppose we pull up the upper plate starting from a plate separation of 0 slowly (i.e., no kinetic energy involved) until a separation of d is reached. See the following figure.

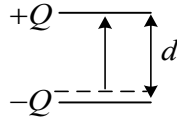


Figure. Setting up the capacitor system by pulling up the upper plate.

During this process, the force on the upper plate is $QE/2$ (Coulomb's force), the total work done is $QEd/2$, where d is the total distance traveled, and $E/2$ accounts for the electric field generated by the bottom plate (treat the bottom plate as the charge generating the field, while the upper plate as the test charge to measure the Coulomb's force), while E is the total field. Note, $Ed = V_0$.

$$U = \underbrace{Q \frac{E}{2}}_F d = (CV_0) \cdot \left(\frac{E}{2} d\right) = \frac{1}{2} CV_0 \cdot V_0 = \frac{1}{2} CV_0^2$$

The energy is uniformly stored within the electric field, so the **electric energy density** can be calculated by dividing the total energy by the total volume, $A \cdot d$:

$$w_E = \frac{\frac{1}{2} CV_0^2}{Ad} = \frac{1}{2} \frac{\epsilon_0 A E^2 d^2}{d Ad} = \frac{1}{2} \epsilon_0 E^2$$

3) Insert a dielectric between the two plates

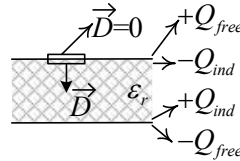


Figure. A parallel plate capacitor with a dielectric material between the two plates.

After inserting a dielectric material with dielectric constant ϵ_r between the two plates, the electric field created by the free charge on the metal plates will induce a thin layer of bound charge on both sides of the dielectric (see the figure above). The bound charge has an opposite polarity to the adjacent free charge layer.

Using the boundary condition for the electric displacement $\vec{D}$ (assume positive direction pointing downward), $D = \sigma_f$, as a result,

$$E = \frac{D}{\epsilon_0 \epsilon_r} = \frac{\sigma}{\epsilon_0 \epsilon_r}, \quad V = Ed = \frac{\sigma d}{\epsilon_0 \epsilon_r}$$

$$C = \frac{Q}{V} = \frac{A\sigma}{\sigma d} \cdot \epsilon_0 \epsilon_r = \left(\frac{A\epsilon_0}{d} \right) \cdot \epsilon_r \quad \leftarrow \text{increased by } \epsilon_r \text{ times}$$

We can also compute the bound charge density on the surface, for example, on the bottom surface (recall, $\sigma_B = \vec{P} \cdot \hat{n}$),

$$\sigma_B = P = \epsilon_0 \chi_e E = \epsilon_0 \chi_e \frac{\sigma_f}{\epsilon_0 \epsilon_r} = \frac{\chi_e}{\epsilon_r} \sigma_f = \frac{(\epsilon_r - 1)}{\epsilon_r} \sigma_f = \left(1 - \frac{1}{\epsilon_r} \right) \sigma_f$$

Example:

Evaluate the capacitance for the following two cases.

Case 1: $A = 1\text{cm}$, $d = 1\text{ mm}$, vacuum between the two plates.

Solution:

$$C = \frac{A\epsilon_0}{d} = \frac{(1\text{cm})^2 \times 8.85 \times 10^{-12} [\text{F/m}]}{1\text{mm}} = 0.9 \text{ pF}$$

Case 2: $A = 25\text{cm} \times 5\text{cm}$, $d = 0.01\text{ mm}$, a dielectric material with $\epsilon_r = 10$ in between.

Solution:

$$C = \frac{(25\text{cm} \times 5\text{cm}) \times 8.85 \times 10^{-12} [\text{F/m}]}{0.01\text{mm}} \times 10 = 11 \mu\text{F}$$

Now you can appreciate that [Farad] is a huge unit. Can you imagine that there is a modern technology called “supercapacitor”, which has a capacitance reaching up to 100~10000 F (with a output voltage of 2.2~3.3V)?

4) A more complex situation

Consider the following capacitor, a gap is left between the dielectric slab and the capacitor plate on both sides. The whole device is in a vacuum.

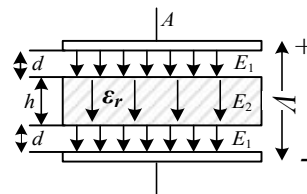


Figure. A parallel plate capacitor with a dielectric slab in the center.

Between the two plates, the electric displacement D should be continuous across the two vacuum-dielectric boundaries due to the lack of free surface charge on these boundaries.

$$E_1 = E, \quad E_2 = \frac{E}{\epsilon_r}$$

$$\Rightarrow Ed + \frac{E}{\epsilon_r} h + Ed = V \Rightarrow E = \frac{V}{2d + h/\epsilon_r}$$

The induced charge density on the dielectric material's top surface (applying Gauss's law in a thin pillar across the boundary, as we have done when deriving the boundary conditions):

$$\underbrace{-E \cdot A}_{\text{top surface}} + \underbrace{\frac{EA}{\epsilon_r}}_{\text{bottom surface}} = \frac{\sigma_{ind} A}{\epsilon_0} \Rightarrow \sigma_{ind} = -\left(1 - \frac{1}{\epsilon_r}\right) \underbrace{\epsilon_0 E}_{\sigma_{free}} = -\left(1 - \frac{1}{\epsilon_r}\right) \sigma_f$$

Now, let's put in some numbers: $d = 1\text{mm}$, $h = 3\text{mm}$, $V = 30\text{KV}$, $\epsilon_r = 5$,

$$\Rightarrow E = 11.5 \left[\frac{\text{MV}}{\text{m}} \right], \quad \sigma_{ind} = 0.08 \left[\frac{\text{mC}}{\text{m}^2} \right], \quad \sigma_{free} = 0.10 \left[\frac{\text{mC}}{\text{m}^2} \right].$$

Now, put the whole system in air and redo the problem.

The air gap cannot sustain a E field as strong as 11.5 MV/m (the electric breakdown field strength is 3 MV/m); rather, free charges will be deposited on the top and bottom surfaces of the dielectric slab through corona discharge, which will effectively reduce the field strength in the airgap.

The maximum field strength in the air gap is $E_{1,max} = 3 \left[\frac{MV}{m} \right]$,

So, the voltage across the airgap will be clamped at $V_1 = 3 \times 10^6 \times 1 \times 10^{-3} = 3 [KV]$

What happens is that due to corona discharge, a thin layer of positive charge (free charge) will be deposited on the top of the dielectric material, while a layer of negative charge is deposited on the bottom surface. See the figure below.

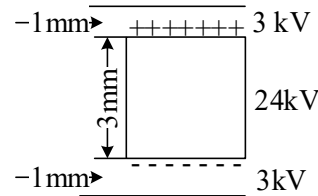


Figure. The top and bottom surfaces of the dielectric slab are coated with free charges through corona discharge.

The total voltage drop between the two plates is:

$$2V_1 + V_{dielectric} = 30KV \Rightarrow \text{The voltage drop on the dielectric slab is } V_{dielectric} = 24KV$$

We are ready to compute the surface free charge density by using the boundary condition for D across the top surface of the slab:

$$\sigma_f = D_2 - D_1 = \epsilon_0 \epsilon_r \left(\frac{V_2}{h} \right) - \epsilon_0 \frac{V_1}{d} = 0.33 \left[\frac{mC}{m^2} \right]$$

The free charge density on the top plate can be computed from the boundary condition for E:

$$\sigma_f = \epsilon_0 E_{1,max} = 0.027 \left[\frac{mC}{m^2} \right]$$

5) Capacitor paradox

Two identical capacitors are connected in parallel. One of the capacitors is charged with a voltage of V_0 , the other is uncharged. When the switch is closed, compute the voltage, charge amount, and energy of the system, and compare with the quantities before closing the switch.

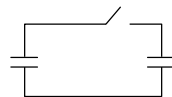


Figure. The two capacitor paradox.

Apparently, after the switch is closed, $Q \rightarrow \frac{Q_0}{2}$, $V \rightarrow \frac{V_0}{2}$,

Total energy: $U_{tot} = \frac{1}{2} \left(\frac{V_0}{2} \right)^2 \times 2 = \frac{CV_0^2}{4} = \frac{1}{2} \left(CV_0^2 \cdot \frac{1}{2} \right) = \frac{1}{2} U_{before}$. Where does the energy go?